

TERS + Machine Learning — Portfolio Summary

This document summarizes results from two complementary pipelines: **(i)** CNN1D on synthetic Raman spectra and **(ii)** Vision Transformer on 2D TERS-like maps.

Overall Metrics

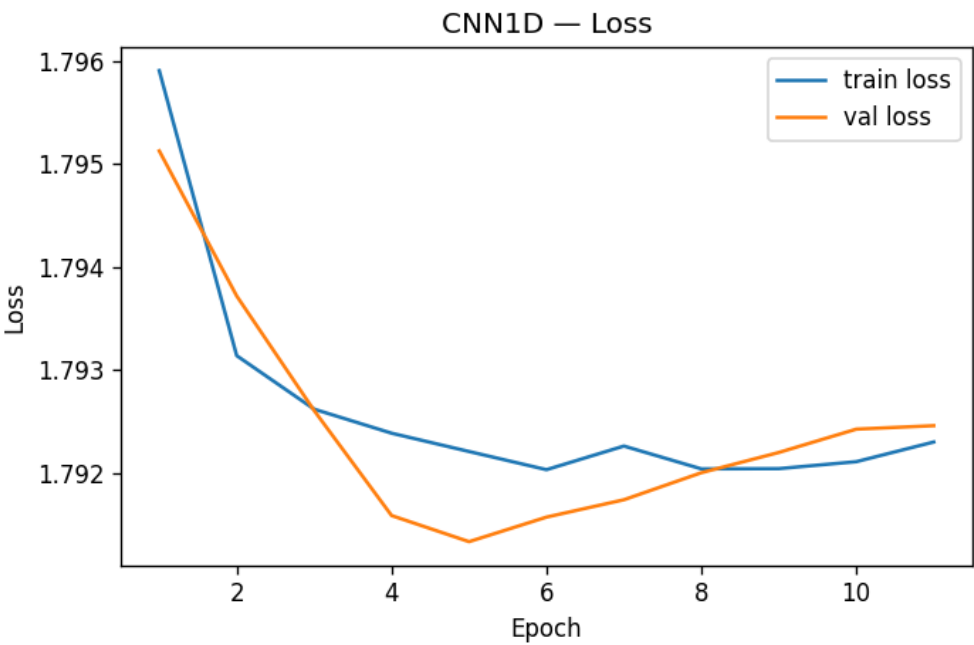
Model	Accuracy	F1-macro
CNN1D (spectra)	0.174	0.049
ViT (TERS maps)	0.164	0.113

Per-class Metrics (Top-8)

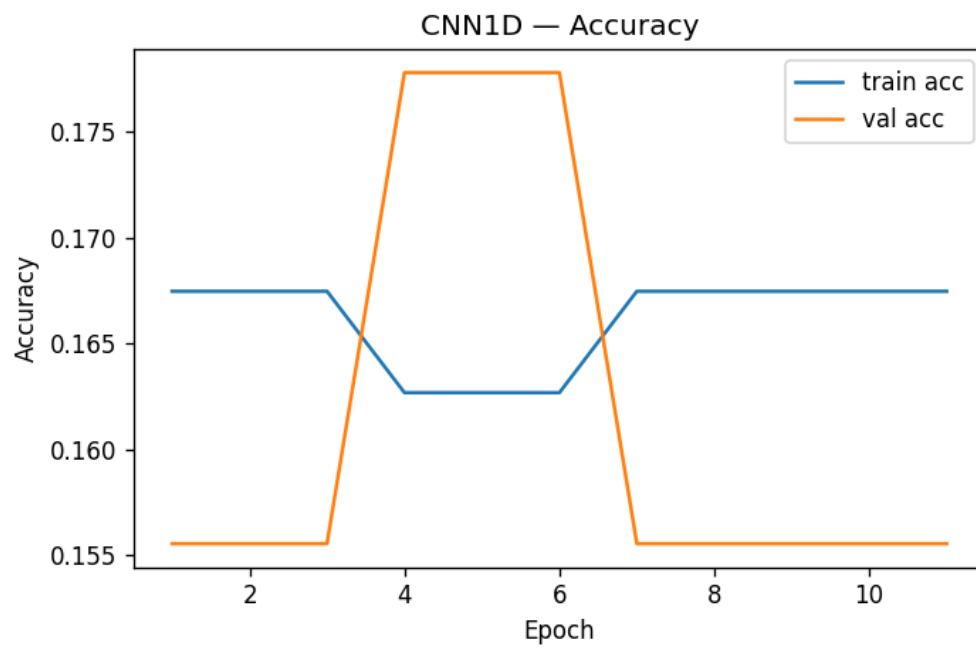
Class	F1 — CNN	F1 — ViT
accuracy	0.174	0.164
anthracene	0.000	0.000
benzene	0.000	0.235
coronene	0.296	0.250
gnf_small	0.000	0.000
macro avg	0.049	0.113
naphthalene	0.000	0.000
pyrene	0.000	0.194

Key Figures

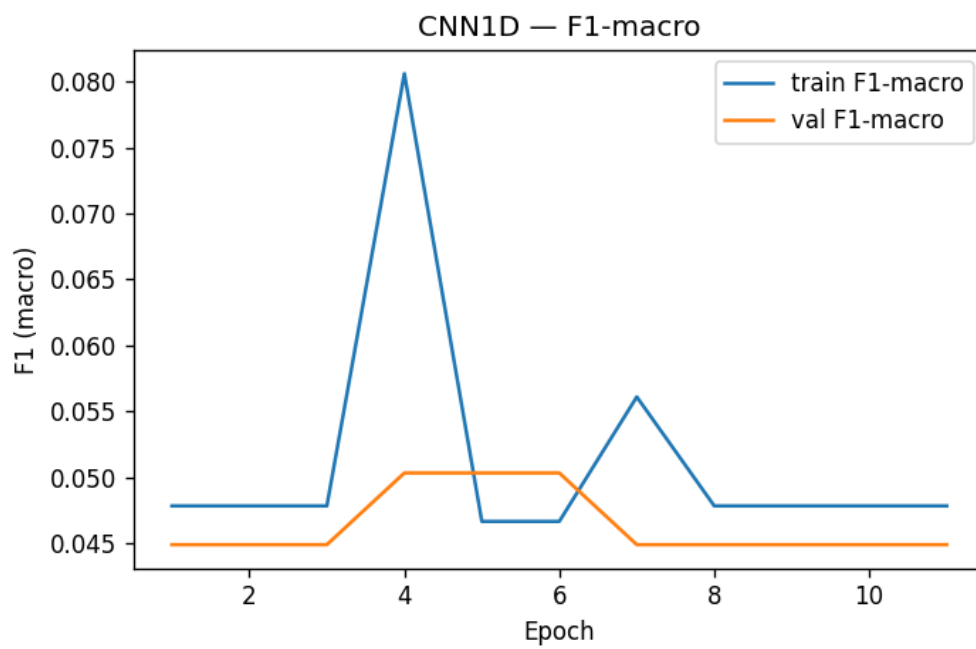
cnn1d_loss.png



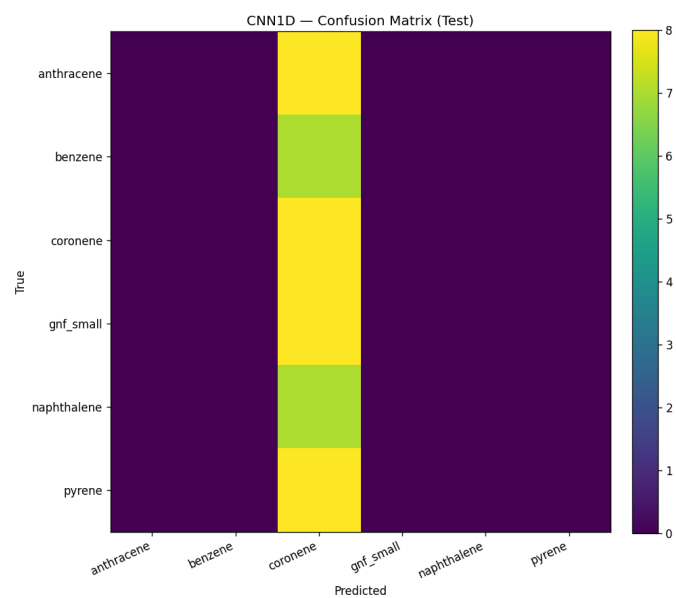
cnn1d_acc.png



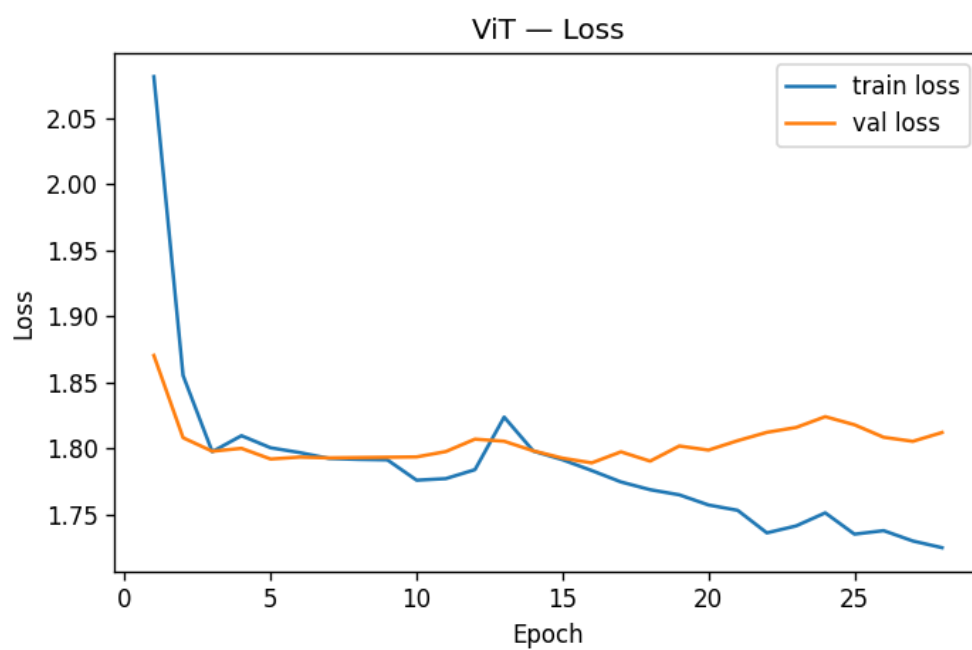
cnn1d_f1.png



cnn1d_confusion_matrix.png



vit_loss.png



vit_acc.png

